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Polymorphism: The Same and Not Quite the Same

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ABSTRACT: Molecular similarity and supramolecular dissimilarity are discussed with reference to the definition of the term *polymorph*.

Introduction

The subject of polymorphism now seizes the attention of a significant number of chemists and crystal engineers associated with the pharmaceutical industry.[1,2] The phenomenon is difficult to understand and more so to predict, but it is of great importance in a legal and commercial sense, and there is a concomitant urgency in addressing pertinent research issues. In this note, I would like to enumerate and briefly discuss some unusual aspects of this (still mysterious[3]) phenomenon: these comments are intended as points for discussion and debate. The reader may not find too many neat answers here. In some ways, progress in our understanding of polymorphism today requires more the framing of correct questions rather than the obtaining of answers.

Difficulties sometimes arise in the very definition of the terms *polymorph* and *polymorphism*. Most currently accepted definitions are based around the idea that it is a phenomenon that involves different crystal structures of the same chemical compound.[4] Considering that the term *polymorph* itself is rather old (Mitscherlich first used it in 1820),[5] it has certainly stood the test of time well. The first of the modern definitions is probably McCrone's which states that *a polymorph is a solid crystalline phase of a given compound resulting from the possibility of at least two different arrangements of the molecules of that compound in the solid state*.[6] The implication here is that there are at least two crystal forms, and that in each of these forms, there is a different arrangement of the molecules of the compound. In other words, there is an implied correspondence between the different molecular arrangements and the different crystal forms. In most cases, this is indeed the case, and there is no further issue. However, ambiguities do crop up: consider, for example, our recent studies on aspirin.[7–9] Here, one has two idealized forms both of which contain practically identical layers of carboxylic acid dimers. The difference between these forms lies in the relative dispositions of the layers. If this were all, there would be no particular problem, and the two forms could uneventfully be termed polymorphs. However, real crystals of aspirin contain intergrowths of both domains in a structural continuum. The matter is further complicated by the fact that only one of the two idealized structures has been realized experimentally. So we have here a molecular solid that contains two different arrangements of the molecules *within the same crystal*. Is this a polymorph? If so, to what crystal is it polymorphic? Would crystals containing different proportions of the two domains (as found experimentally[8]) be called different polymorphs? In the end, how many polymorphs of aspirin exist: one, two, or infinite? All three answers would seem to be plausible, depending on one's point of view. It could be said that there is only one polymorph, and that all the crystals are structural modulations. Alternatively, one could say that there are two polymorphs corresponding to each of the idealized forms. Finally, one could also maintain that there are an infinite number of polymorphs corresponding to different domain ratios of the two idealized forms. To summarize, it is clear that conventional definitions of polymorphism are untenable for aspirin: they are too limited in their scope.

Polymorphs are *different* crystal forms of the *same* chemical compound or substance. Conventional wisdom suffices as long as one is confident about how different two crystal structures need to be to term them different, and how similar two chemical compounds need to be to call them the same. In many cases, there is indeed no ambiguity on either of these counts, and it is

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easy to determine if two given crystals are polymorphic or not. But there are borderline cases where this matter is not at all clear.

Let us first consider differences in crystal structure. When are two structures different? Crystals may display different metric features or different symmetry, or both. When two structures have different crystal lattices, one may term them as being crystallographically distinct. What about cases when one or more cell edges are increased by integral multiples corresponding to superstructures? Many such structures are being detected today with CCD diffractometers from weak spots in the X-ray diffraction patterns, so that one obtains cells wherein one of the axes may be, say, doubled or tripled. The choices of unit cell are surely different, but are these really "different" structures? Chemically speaking, I would venture to say that many such structures are the "same". It would be very hard to imagine a situation where two such structures have distinct properties and presumably patentability. Much the same situation pertains to those structures with multiple Z' values wherein formally symmetry-independent molecules are related by some kind of pseudosymmetry that verges toward a higher symmetry structure with a lower value of Z'. Improved methods of data collection and structure analysis might result in pairs of such structures being called polymorphs. This might be technically correct, but do these structures qualify for the polymorph label, in spirit? In an interesting sidelight, two structures of a fused ring β-lactam have been reported wherein the unit cells, space groups, and positioning of the hydrocarbon residues (which constitute the major fraction of the molecule) are practically the same; what is different is the orientation of the NH–CO fragments, and the orientations of the resulting hydrogen bonds.[10] Most chemists would not have a problem in terming these structures as polymorphs, based on the chemical differences in the hydrogen bonding in the two forms, but the formal crystal structures and overall packing are exceedingly similar. So, when it is a matter of chemical common sense versus quantitative crystallographic indicators, which criterion prevails? As a postscript to this line of thought, it may be added that one's ability to discern differences in crystal structure, in these borderline cases, also seems to depend on the technical capabilities of the available instrumentation. Since this will vary with time, will the definitions of what is and what is not a polymorph also be time dependent?

More vexing are some situations in which one is asked to assess the similarity of two molecular structures. Are tautomers the same or different molecules? If tautomers represent the "same" molecule, a crystal structure with a particular tautomer is polymorphic to a different crystal structure that contains another tautomer. If they are different, the so-called tautomeric polymorphs are not polymorphs at all but crystals of different compounds. We encountered an intriguing situation in the crystal structure of the drug omeprazole in which crystals contain variable amounts of two tautomers (the so-called 5-methoxy and 6-methoxy derivatives).[11] The tautomeric compositions may be established with X-ray crystallography, and a range of compositions is seen. The crystal packing in both tautomers is the same, and any crystal may be modeled as a solid solution of the 5-OMe tautomer in the 6-MeO tautomer. As in aspirin, there is a structural continuum, but this time it occurs at the molecular level. Are these crystal structures different? Are the molecular structures the same? We note that the photochemical properties of some of these crystals are different: three particular compositions even enjoy distinctive patent protection on this basis.[12,13] Are they polymorphs? If so, how many polymorphs of omeprazole exist—one, two, or infinite? Should definitions of polymorphism rely so heavily on structural criteria? Buerger defined polymorphs as crystal forms with different properties.[3] Would properties be a more realistic and less ambiguous indicator of polymorphism than structure? Are subtle structural differences really meaningful especially in the context of the kind of modulation that is seen in omeprazole and aspirin? Should one assess molecular sameness with a certain flexibility and latitude? Accordingly, how important is the criterion that the molecular structure should be *exactly* the same if two crystals are to be called polymorphs? In an opposite vein, it is known that there are structures of different but isometric molecules, with essentially the same crystal structure. This phenomenon has been termed *synthomorphism* because of the local similarity in the crystal structures.[14] Other terms that have been used in the field are isomorphism, isotypism, isostructuralism, homostructuralism, and even qualifying epithets, for example *approximate isomorphism*. This plethora of definitions is clearly indicative of an impasse—one cannot define or limit a complex scenario such as a crystal structure with overly simple and therefore inflexible criteria. The definitions themselves must become looser so that some of them merge.

The idea of a structural landscape[15] that includes a number of structural variations of the same molecular species is one such loosening and is an attractive proposition in the present context. Within this idea is the accompanying notion that by the term *molecular species* one means a certain molecule and also the chemical space around it that is populated by salts, co-crystals, solvates, hydrates, and, if one were to become very flexible, even some substituted derivatives of the parent molecule.[16] These variations in the basic molecular structure are associated with certain crystal structures. Then, if any structure in the landscape within an agreed window of molecular similarity (not just the *exact* structural formula of a particular molecule) has a crystal structure that is beyond some agreed criterion of distinctiveness from the others, then such crystal structures could be termed polymorphs. If not, other terms might be required such as *pseudopolymorph* which represents a crystal form of a species within the landscape which falls beyond an accepted window of molecular similarity. Conformational isomers could also exist comfortably within a structural landscape as would salts and multicomponent crystals. The matter could then be reduced to a discussion of what are acceptable windows for molecular similarity and supramolecular dissimilarity; the sceptic would argue that these are the very issues that bedevil the subject today, but the advantage in accepting the idea of the structural landscape is that chemically reasonable arguments might find a place in framing definitions and conventions.

It is clear that polymorphism will continue to remain in the forefront of pharmaceutical crystal engineering for some time to come.[17–19] Polymorphs are very special extensions of pharmaceutical space around a particular drug molecule, because they can exist not only for the drug molecule itself but also for other extensions of pharmaceutical space such as co-crystals, salts, and solvates. In other words, one can obtain polymorphs of salts, polmorphs of co-crystals, and so on. To conclude, it might seem that the questions posed above amount to a mere quibbling over words. However, a lot of time and money is being spent on polymorph research today, and legal and regulatory decisions with considerable financial implications are being taken on the basis of these ongoing research developments.[20,21] Accordingly, even seemingly peripheral issues need to be addressed, scrutinized, and evaluated diligently. It is in this spirit that this short comment has been written.

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